

Predictive Analytics for Smart Electric Vehicle Charging Station Recommendation

1. Project Overview

Electric Vehicle (EV) charging demand is increasing rapidly, making it important to understand charging behavior and recommend the best charging station for users. This project analyzes **1,300 EV charging sessions** using Python and Machine Learning. It performs Exploratory Data Analysis (EDA), statistical analysis, feature engineering, and predictive modeling to build a **Smart EV Charging Advisor** that recommends the most suitable charging station based on historical charging patterns.

2. Data Source

Source: Gigasheet Dataset – EV Charging Patterns

Dataset Size: Approximately 1,320 charging session records more than 20 features

Location: United States

Timeline: 2024

Domain: Electric Vehicles (EV), Energy Management, and Transportation Analytics

3. Problem Statement

The increasing adoption of Electric Vehicles (EVs) has led to a higher demand for efficient and accessible charging infrastructure. However, EV users often face challenges in selecting the most suitable charging station due to variations in charging time, charging cost, charger type, energy consumption, and battery status. Without data-driven insights, users may experience longer waiting times, higher charging expenses, and inefficient charging decisions.

This project addresses these challenges by analyzing historical EV charging data and applying Machine Learning to predict charging duration and recommend the most appropriate charging station. The proposed Smart EV Charging Advisor helps users make informed charging decisions by estimating charging time, charging cost, and providing intelligent station recommendations based on charging patterns and historical data.

4. Objectives

Analyze EV Charging Patterns

To explore and understand electric vehicle charging behaviour using exploratory data analysis (EDA).

To identify trends in charging duration, charging cost, energy consumption, and charging rate.

Study Energy Consumption and Charging Performance

To examine the relationship between battery capacity, energy consumed, charging duration, and charging rate.

To identify the factors that influence charging efficiency.

Evaluate Charging Cost Patterns

To analyze charging costs across different charging stations, charger types, and time periods.

To identify cost-efficient charging options for EV users.

Compare Charging Station Performance

To compare charging stations based on charging duration, charging cost, charging rate, and energy delivered.

To identify high-performing charging stations

Identify Factors Affecting Charging Duration

To investigate how charging duration is influenced by battery gain, charging rate, battery capacity, and environmental temperature.

Predict Charging Duration Using Machine Learning

To build and evaluate a Random Forest Regression model that predicts charging duration based on charging session characteristics.

5. Attribute (Column / Feature) Details

User ID – A unique identifier assigned to each EV user.

Vehicle Model – The model of the electric vehicle used.

Battery Capacity – The total battery capacity of the vehicle, measured in kilowatt-hours (kWh).

Charging Station – The location or name of the charging station where the vehicle was charged.

Charging Start Time – The date and time when the charging session began.

Charging End Time – The date and time when the charging session ended.

Charging Duration – The total time taken to complete the charging session.

Energy Consumed – The amount of electrical energy consumed during charging, measured in kWh.

Charging Cost – The total cost incurred for the charging session.

Charging Rate – The speed at which the vehicle was charged.

State of Charge (Start) – The battery percentage before charging started.

State of Charge (End) – The battery percentage after charging was completed.

Distance Driven – The distance traveled by the vehicle before charging.

Temperature – The ambient temperature during the charging session.

Vehicle Age – The age of the electric vehicle.

Charger Type – The type of charger used, such as Fast Charger or Slow Charger.

User Type – The category of the EV user (for example, private or commercial user).

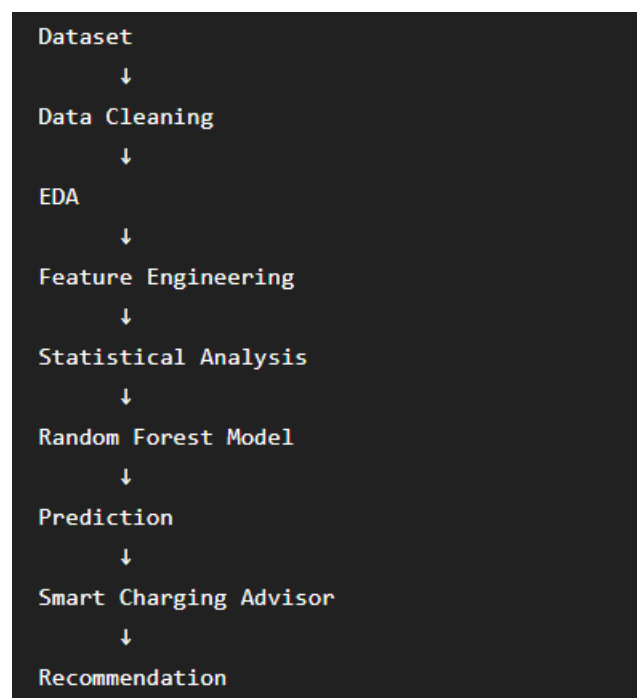
Time of Day – The period during which charging occurred, such as Morning, Afternoon, Evening, or Night.

Day of Week – The day on which the charging session took place.

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab
- Machine Learning(Random Forest Regression)

7. Project Workflow



8. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using data preprocessing techniques.

Data Cleaning Steps

- Checked data types of all columns
- Identified missing values
- Handled null values
- Removed duplicate records
- Corrected inconsistent data
- Converted data types where required

Feature Engineering

Feature engineering was carried out to create new useful features from the existing data and improve the performance of the machine learning model.

Feature Description

Battery Gain-Created a new column by calculating the increase in battery percentage after charging (End State of Charge – Start State of Charge).

Vehicle Age Category-Converted vehicle age into categories such as New, Medium, and Old for easier analysis.

Charging Duration-Used the charging start time and end time to calculate the total charging duration (in hours).

Time of Day-Used the existing time information to analyze charging behavior during different times of the day (Morning, Afternoon, Evening, Night).

Encoded Categorical Data-Converted text columns like Charger Type, User Type, and Vehicle Model into numerical values so they could be used by the machine learning model.

Feature Selection-Selected important features such as battery capacity, charging rate, energy consumed, charging cost, charger type, and battery gain for model training.

8. Exploratory Data Analysis (EDA)

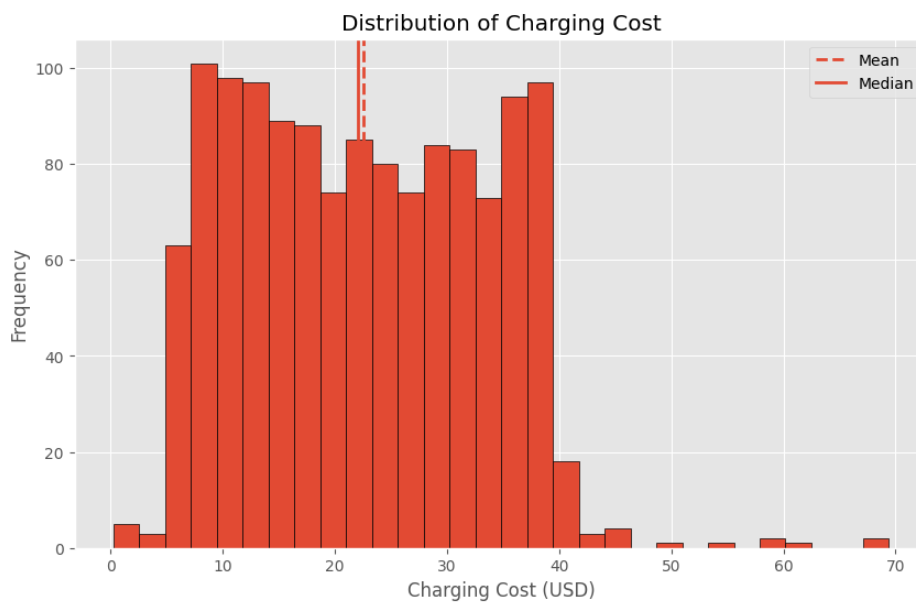
- Data frame structure verification
- Statistical summary of numerical columns
- Null value verification
- Duplicate record validation

EDA helped in understanding the distribution and characteristics of the data.

10. Statistical Analysis

Measures included:

- Mean
- Median
- Mode
- Minimum value
- Maximum value
- Standard deviation
- These measures helped understand the central tendency and variability of the data.



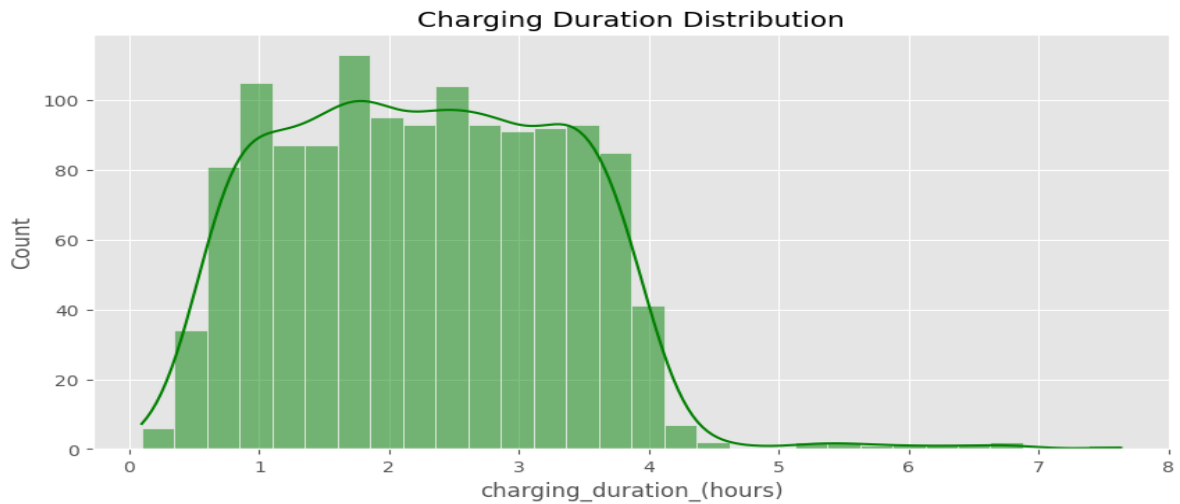
Interpretation

- The majority of charging sessions cost between
 - The mean and median are both around \$22, indicating the average charging cost is close to the middle value and the distribution is fairly balanced.
 - A few charging sessions have costs above
 - Most users incur moderate charging costs, while only a small percentage experience unusually high charging expenses.
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11. Data Visualization

1. Univariate Analysis

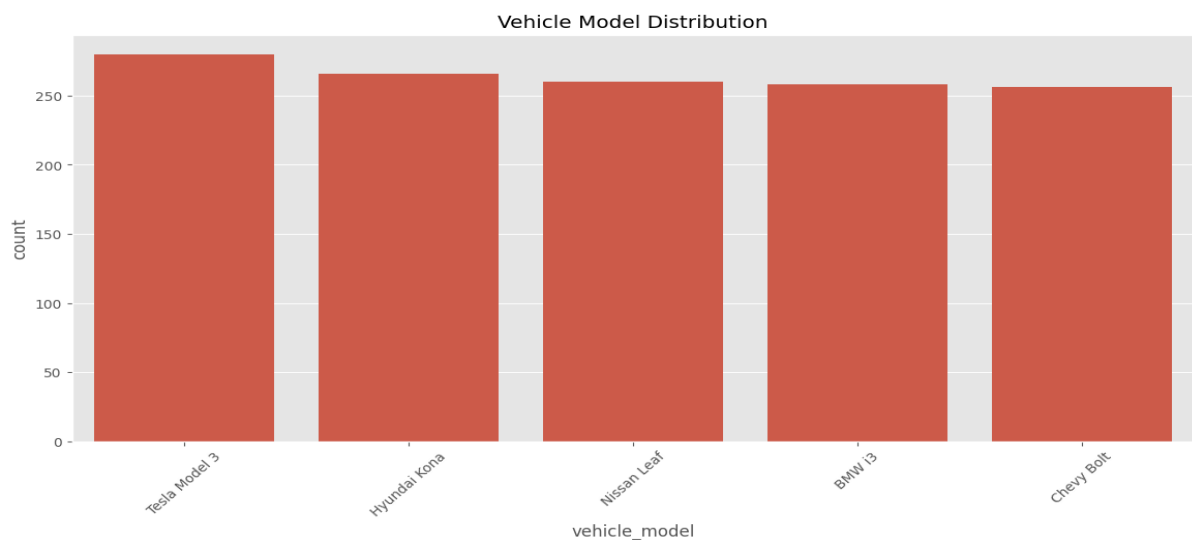
Charging Duration



Interpretation

- Most charging sessions last between 1 and 4 hours, with the highest frequency around 1.5–3.5 hours.
- Very short charging sessions (less than 0.5 hours) are uncommon.
- Charging sessions longer than 5 hours are rare and represent only a small number of observations.
- The distribution is slightly right-skewed, indicating that while most charging sessions finish within a few hours, a few sessions take significantly longer.

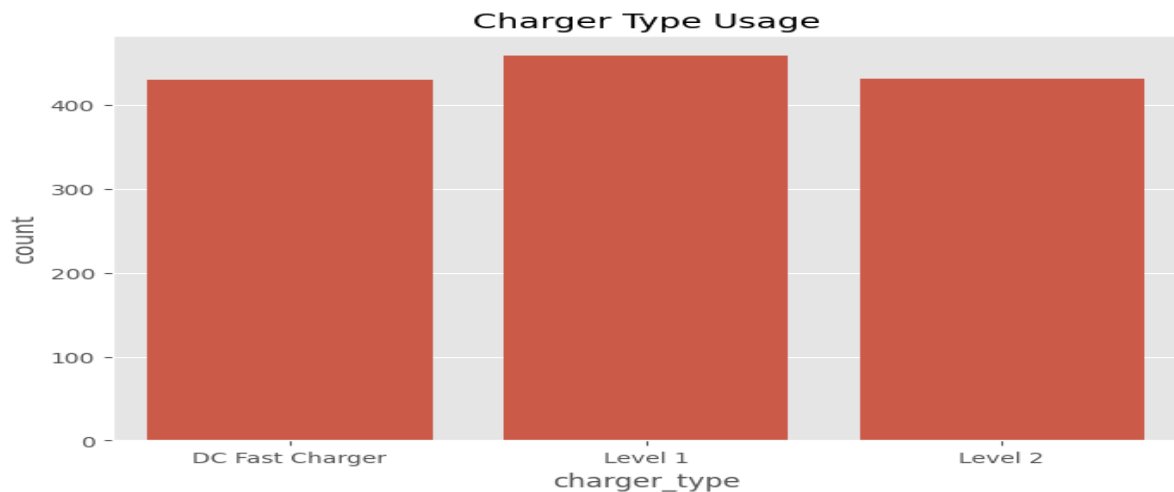
Vehicle Model Analysis:



Interpretation

- Tesla Model 3 has the highest number of charging sessions with approximately 280 records.
- Hyundai Kona follows with around 266 records.
- Nissan Leaf, BMW i3, and Chevy Bolt each have approximately 258–260 records.

Charger Type

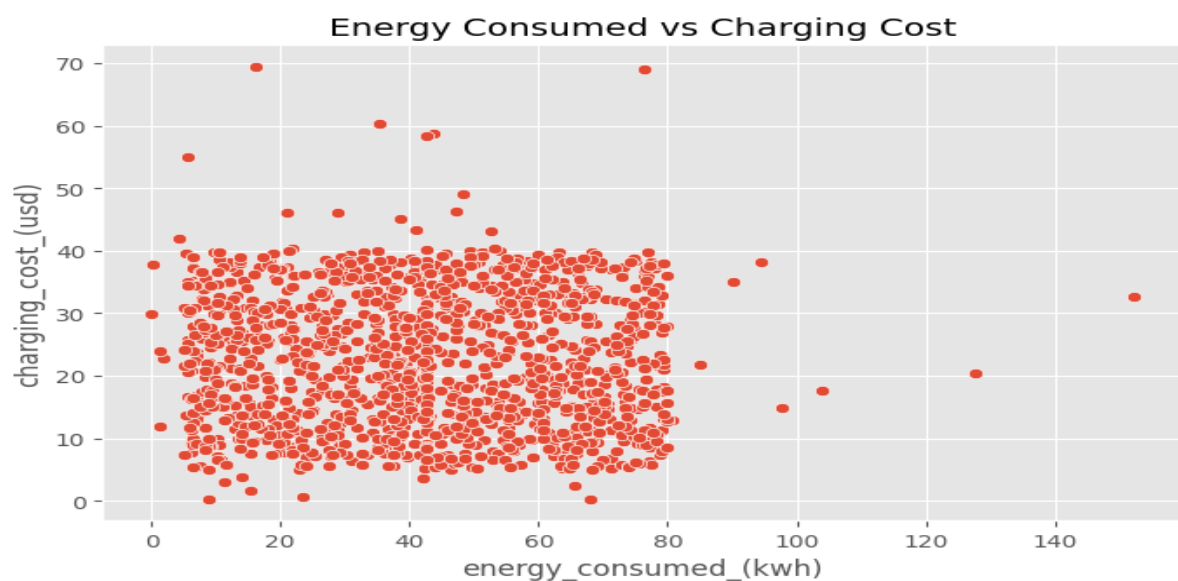


Interpretation

- Level 1 chargers are the most frequently used, with approximately 460 charging sessions.
- DC Fast Chargers and Level 2 chargers each have approximately 430 charging sessions.
- The difference between charger types is relatively small, indicating that users utilize all charger types regularly.

2. Bivariate Analysis

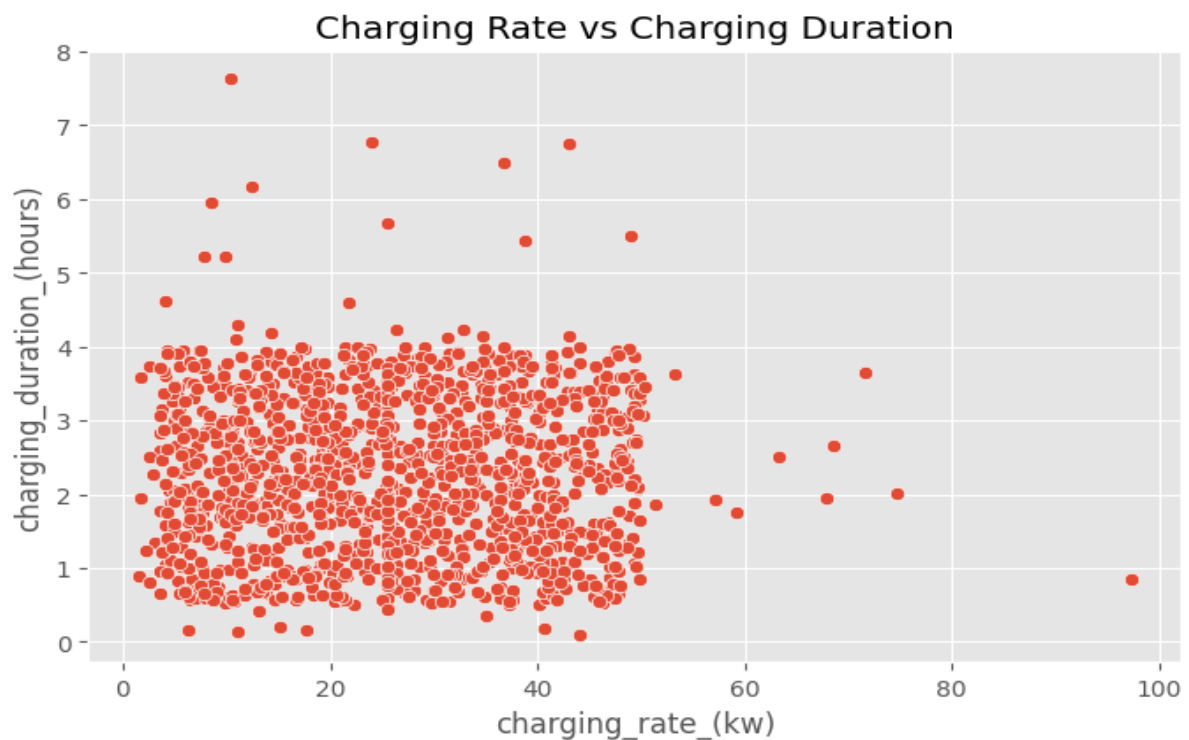
Charging Cost vs Energy Consumed



Interpretation

- Most charging sessions consume between 5 and 80 kWh, while charging costs generally range from
- As energy consumption increases, charging cost generally increases, indicating a positive relationship between the two variables.
- A few sessions consume more than 100 kWh, but these are rare compared to the majority of observations.

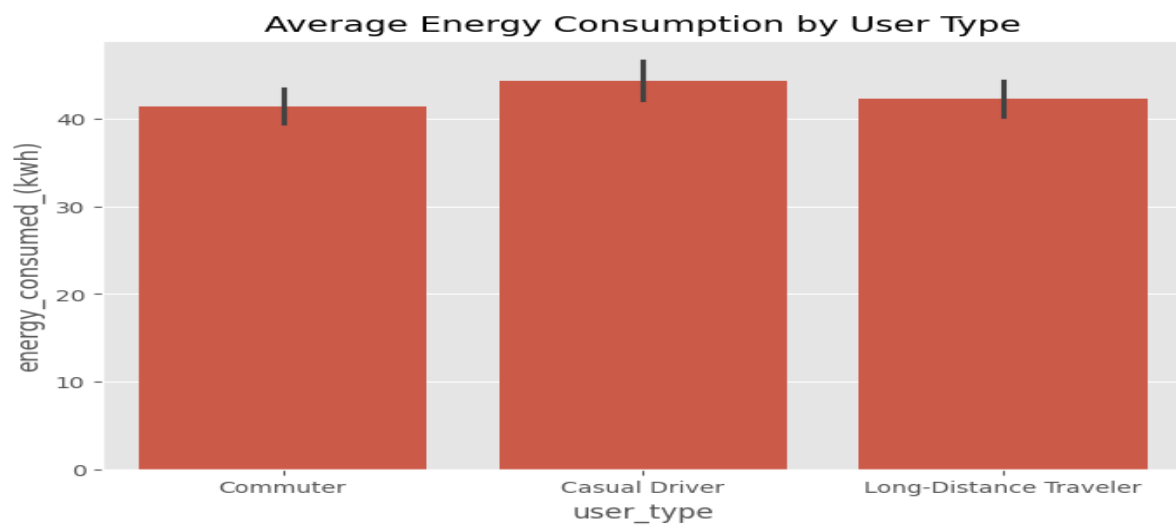
Charging Rate vs Charging Duration



Interpretation

- Most charging rates lie between 5 and 50 kW, while charging durations are typically between 1 and 4 hours.
- Higher charging rates are generally associated with shorter charging durations, showing an inverse relationship.
- A small number of charging sessions last more than 6 hours, representing unusual or slow charging events.
- Charging rates above 60 kW are less common and usually result in charging durations below 3 hours.

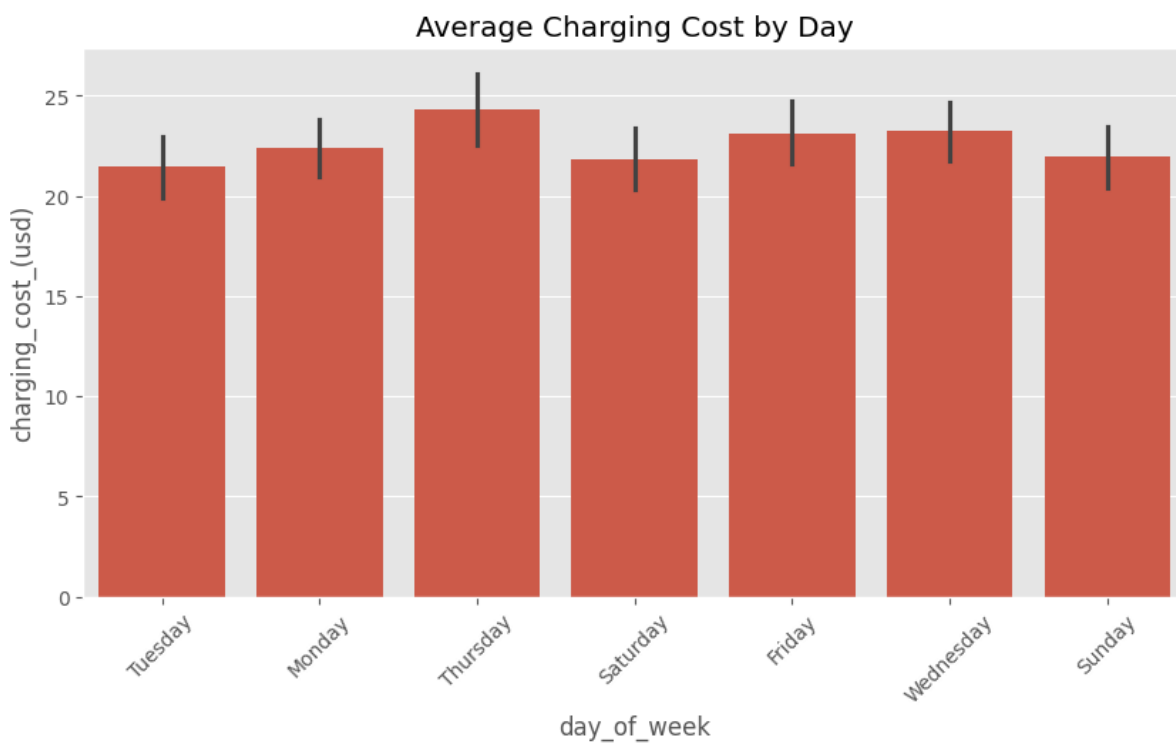
User Type vs Energy Consumed



Interpretation

- Casual Drivers have the highest average energy consumption at approximately 44–45 kWh.
- Long-Distance Travelers consume around 42–43 kWh on average.
- Commuters have the lowest average energy consumption at approximately 41–42 kWh.
- The difference between user groups is relatively small (around 3–4 kWh), indicating that energy consumption is fairly consistent across user types.

Day of Week vs Charging Cost



Interpretation

- Thursday has the highest average charging cost at approximately
 - Tuesday has the lowest average charging cost at around
 - The remaining days (Monday, Wednesday, Friday, Saturday, and Sunday) have average charging costs between
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3. Multivariate Analysis

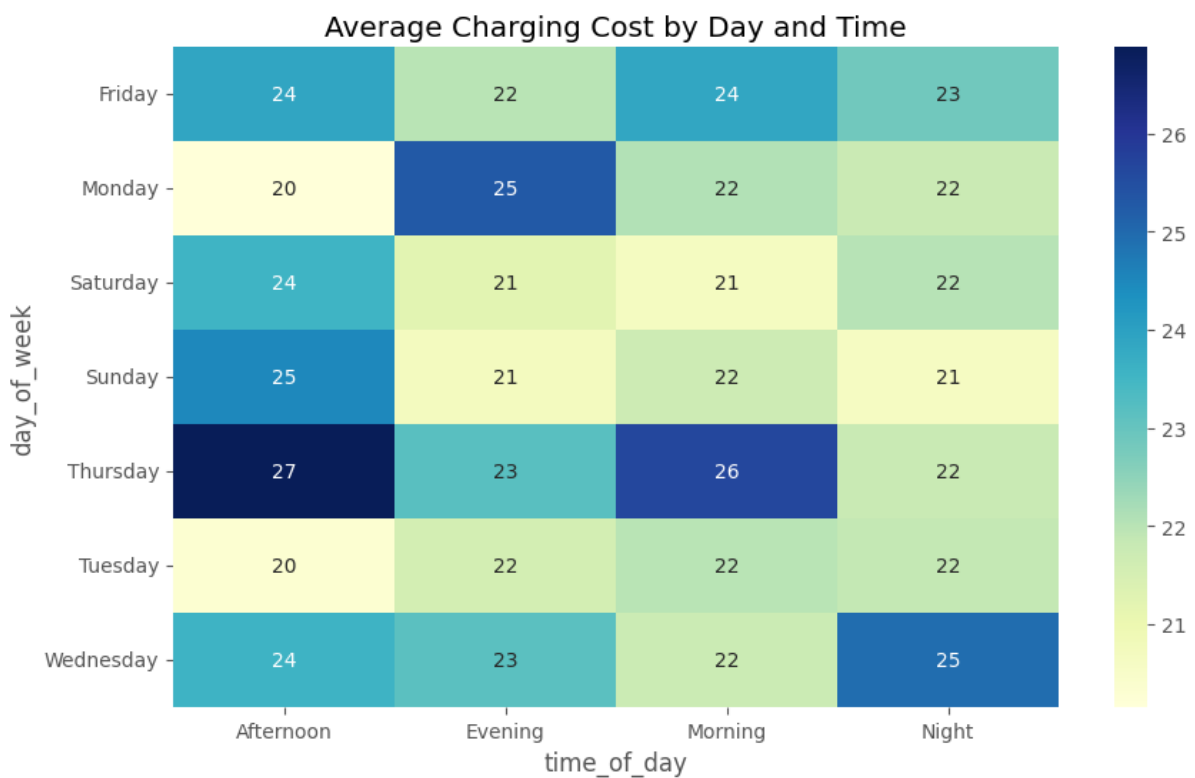
Analysis Performed: Pair Plot



Interpretation

- Most charging sessions have a charging cost between
- The scatter plots show no strong linear relationship between most variable pairs, indicating that charging behavior is influenced by multiple factors rather than a single variable.
- A few outliers are present, such as energy consumption above 100 kWh, charging costs around \$70, charging rates close to 100 kW, and charging durations above 7 hours. These represent rare charging sessions.

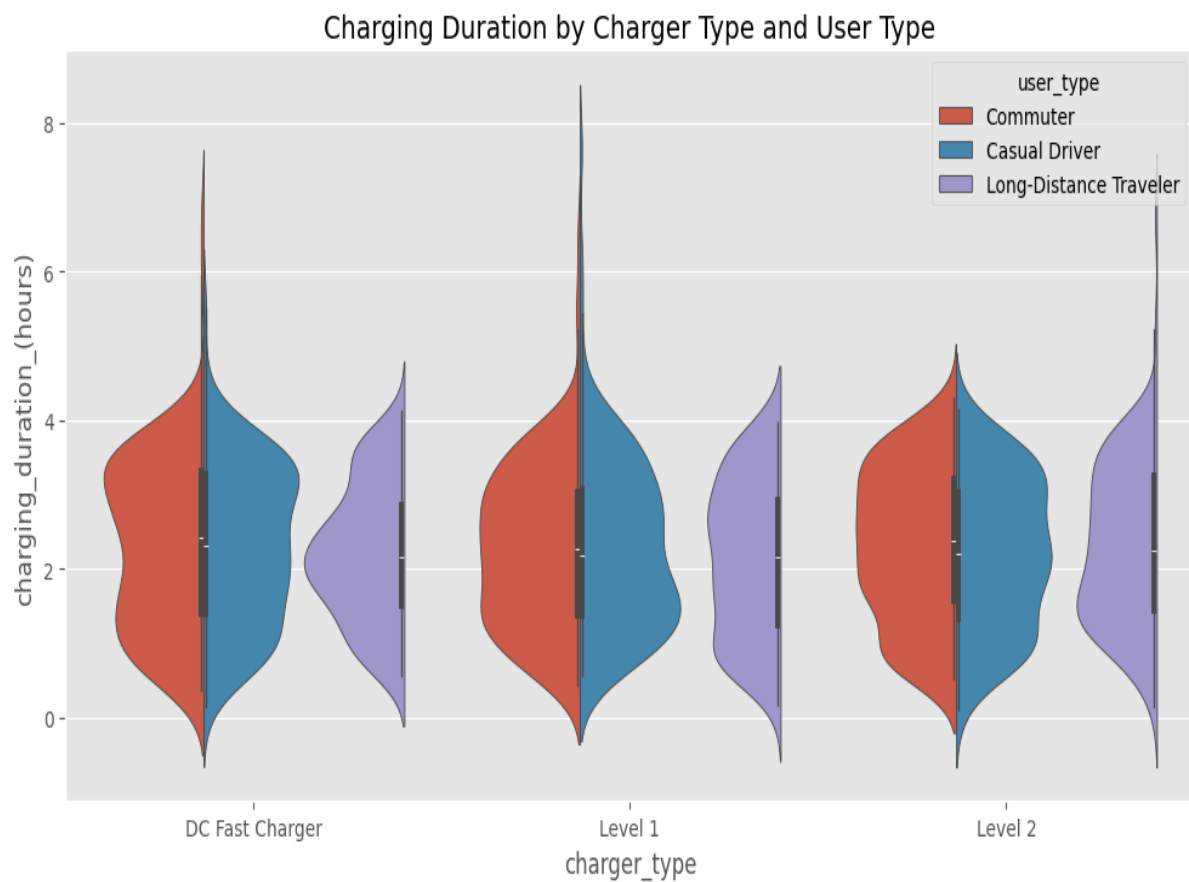
Heat Map



Interpretation

- Thursday Afternoon records the highest average charging cost at approximately 27usd, followed by Thursday Morning at around \$26.
- Monday Evening and Wednesday Night also show relatively high average charging costs of approximately \$25.
- The lowest average charging costs (around \$20) occur on Monday Afternoon and Tuesday Afternoon.
- Most other day-and-time combinations have average charging costs between

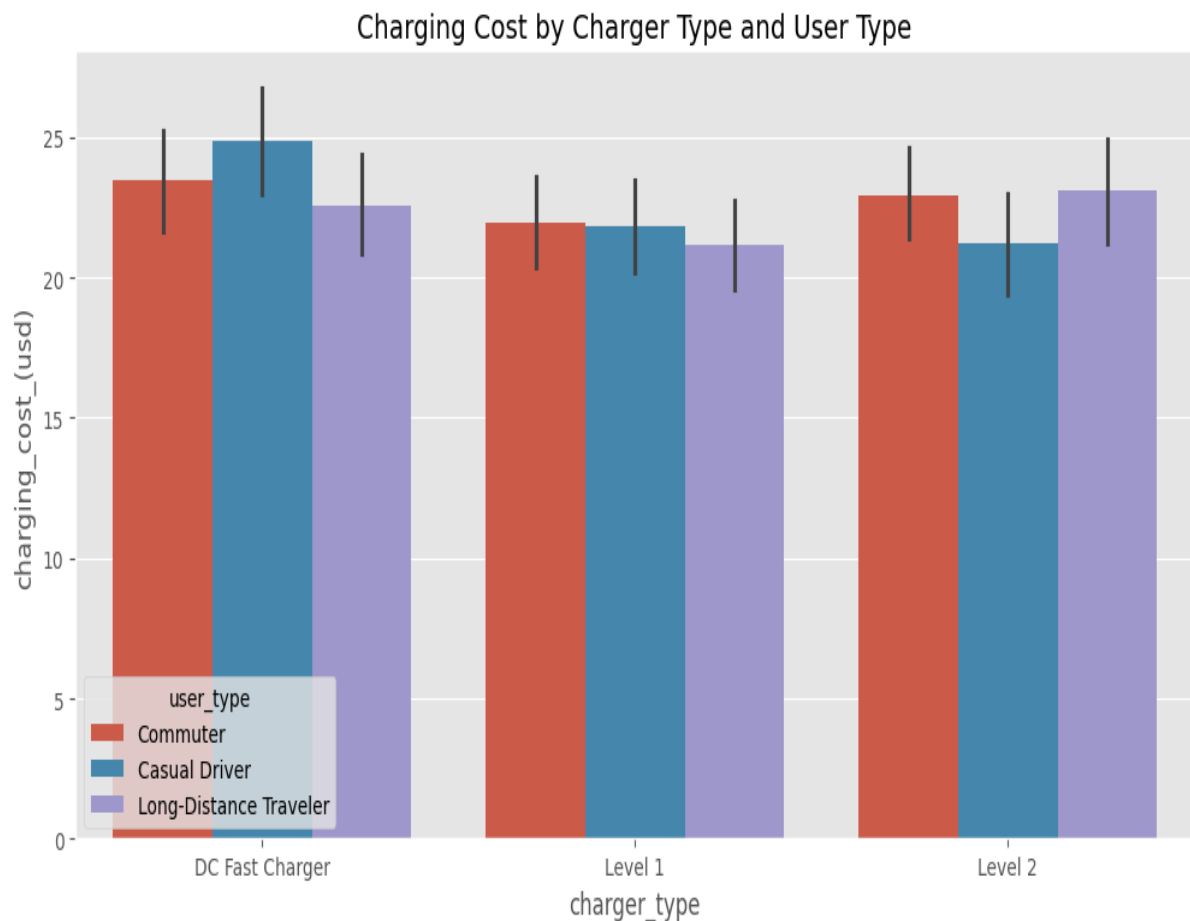
Violin Plot



Interpretation

- **Typical Charging Time (2 to 3 Hours):** Across all charger types and user categories, the main body (widest part) of the distributions is concentrated between 2 and 3 hours, indicating that most users complete their charging session within this timeframe.
- **Maximum Duration Peaks (Up to 8.5 Hours):** Some sessions take significantly longer, reaching maximum peaks of around 7.5 to 8.5 hours (visible in the long upper tails), representing extended or overnight charging sessions.
- **Similar Patterns for Commuters & Casual Drivers:** Both Commuters (red) and Casual Drivers (blue) show almost identical wide distributions, with median charging durations around 2.2 to 2.5 hours.
- **Shorter Sessions for Long-Distance Travelers:** Long-Distance Travelers (purple) generally spend less time plugged in, with their sessions topping out at around 4.5 to 5 hours and a median duration closer to 2 hours.

Grouped Bar Chart

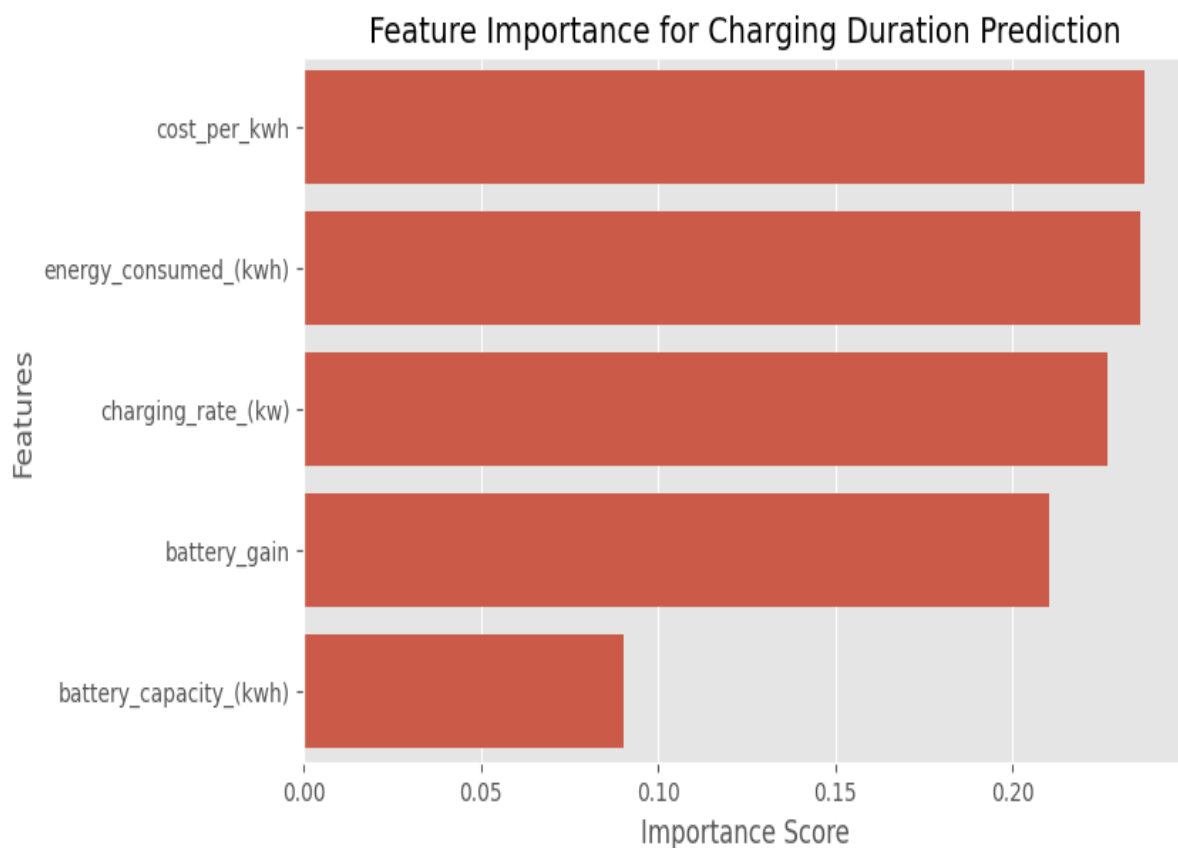


Key Insights & Interpretation

- Consistent Average Cost (21to 25): Across all charger types and user categories, the average charging cost per session remains tightly clustered between 21.00and 25.00.
 - Highest Cost on DC Fast Chargers (~ 25.00):CasualDriversusingDCFastChargersincurthehighestaveragecostatapproximately 25.00, followed by Commuters at roughly \$23.50, reflecting higher premium rates for rapid charging.
 - Lowest Cost on Level 1 Chargers (~ 21.00– 22.00): Level 1 chargers offer the lowest average expenses, ranging from 21.00forLong–DistanceTravelersto 22.00 for Commuters and Casual Drivers.
 - Level 2 Cost Differences (~ 21.20– 23.20): For Level 2 chargers, Long-Distance Travelers pay the most at around \$23.20, Commuters average around 23.00, and Casual Drivers spend the least at about 21.20usd
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12. Machine Learning (Demand Prediction Analysis)

Plot Feature Importance



Interpretation

- **Top Predictor – Cost per kWh (~0.24 / 24%):** The unit cost of energy (`cost_per_kwh`) is the most influential feature in predicting charging duration, with an importance score of approximately 0.24.
- **Strong Influence of Energy & Charging Speed (0.21 to 0.235):** Both Energy Consumed (0.235) and Charging Rate (0.215) play major roles, indicating that total power delivered and charger speed heavily dictate how long a session lasts.
- **Significant Impact of Battery Gain (~0.21 / 21%):** Battery Gain holds an importance score of around 0.21, reflecting that the total charge added during a session directly correlates with time spent plugged in.
- **Lowest Impact – Battery Capacity (~0.09 / 9%):** Total Battery Capacity has the lowest influence among the top features at roughly 0.09, showing that overall battery size matters much less than actual energy transferred and charging rates during a specific session.

13.SMART CHARGING ADVISOR

```
Enter Vehicle Model (e.g., 'Tesla Model 3', 'Hyundai Kona'): Tesla Model 3
Enter Current Battery (%): 20
... Enter Time of Day (Morning/Afternoon/Evening/Night): Evening
Enter Preferred Charger Type (e.g., 'DC Fast Charger', 'Level 1', 'Level 2'): DC Fast Charger
Distance Driven Since Last Charge (km): 70

Analyzing...

=====
🚗 SMART CHARGING ADVISOR
=====

🚗 Vehicle : Tesla Model 3
🔋 Current Battery : 20%
⚡ Energy Required : 59.04 kWh

📍 Recommended Charging Station
Station_167
New York

🕒 Estimated Charging Duration : 1.56 Hours
💰 Estimated Charging Cost : $5.98
👍 Recommendation Score : 100.0/100

Why this station?
✓ Lowest predicted charging time (relative to other options)
✓ Lower charging cost (relative to other options)
✓ High charging rate (relative to other options)
✓ Matches your preferred charger type (DC Fast Charger)
```

14. Summary of Findings

- Most EV charging sessions cost between
 - Most vehicles complete charging within 1–4 hours, making this the typical charging duration.
 - Tesla Model 3 has the highest number of charging sessions, although all vehicle models are represented fairly evenly.
 - Los Angeles records the highest charging station usage, while Chicago has the lowest usage among the analyzed cities.
 - Level 1 chargers are used slightly more frequently than DC Fast Chargers and Level 2 Chargers.
 - Charging costs are generally higher during Thursday Afternoon, while Monday Afternoon and Tuesday Afternoon are the least expensive charging periods.
 - Higher charging rates generally reduce charging duration, improving charging efficiency.
 - Machine learning successfully predicts charging duration, enabling the development of a charging recommendation system.
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15.Types of Analysis

1. Descriptive Analysis

Objective 1: Analyze EV Charging Patterns

Purpose: To explore EV charging behavior and identify trends in charging duration, charging cost, energy consumption, and charging rate.

Metric Result

Total Charging Records 1,300

Average Charging Cost $\approx$ \$22

Average Charging Duration $\approx$ 2.3 Hours

Most Common Charging Cost

Most Common Charging Duration 1–4 Hours

Most Common Energy Consumption 10–80 kWh

Most Common Charging Rate 5–50 kW

Findings

- Most charging sessions cost between
- Around 90% of charging sessions are completed within 1–4 hours.
- Most vehicles consume 10–80 kWh of energy per charging session.
- Charging rates are concentrated between 5 and 50 kW, indicating standard charging conditions for most users.

2. Diagnostic Analysis

Objective 2: Study Energy Consumption and Charging Performance

Purpose: To examine relationships between battery capacity, energy consumed, charging duration, and charging rate.

Analysis Result

Energy Consumed 10–80 kWh for most sessions

Charging Rate 5–50 kW for most sessions

Charging Duration 1–4 Hours for most sessions

High Energy Sessions Above 100 kWh (Rare)

Maximum Charging Duration $\approx$ 7.7 Hours Findings

Charging cost increases as energy consumption increases, indicating a positive relationship.

Higher charging rates generally result in shorter charging durations.

Most charging sessions remain within normal operating ranges, while only a few extreme sessions exceed 100 kWh or 7 hours.

The pair plot shows no single variable alone determines charging behavior, suggesting multiple factors influence charging performance.

3. Diagnostic Analysis

Objective 3: Evaluate Charging Cost Patterns

Purpose: To analyze charging costs across charging stations, charger types, and time periods.

Category Result

Average Charging Cost $\approx$ \$22

Highest Average Cost Thursday Afternoon ($\sim$ \$27)

Lowest Average Cost Monday Afternoon & Tuesday Afternoon ($\sim$ \$20)

Cost Variation During Week

Typical Charging Cost

Findings

- The average charging cost is approximately \$22.
- Thursday Afternoon is the most expensive charging period with an average cost of approximately \$27.
- Monday Afternoon and Tuesday Afternoon are the least expensive periods with average costs of approximately \$20.
- The overall variation in charging cost throughout the week is relatively small (about \$7).

4. Diagnostic Analysis

Objective 4: Compare Charging Station Performance

Purpose: To compare charging stations based on charging duration, charging cost, charging rate, and energy delivered.

Station Approximate Usage

Los Angeles -298 Sessions

San Francisco -265 Sessions

Houston -263 Sessions

New York -255 Sessions

Chicago -243 Sessions

Findings

- Los Angeles has the highest charging station usage with approximately 298 charging sessions.
- Chicago has the lowest station usage with approximately 243 charging sessions.

- The difference between the most and least used charging stations is only about 55 sessions, indicating balanced station utilization.
- Charging stations across all locations exhibit relatively consistent charging performance.

5. Diagnostic Analysis

Objective 5: Identify Factors Affecting Charging Duration

Purpose: To determine how charging duration is affected by battery gain, charging rate, battery capacity, and temperature.

Factor Observation

- Charging Rate-Higher charging rate reduces charging duration
- Battery Gain Larger battery gain increases charging duration
- Battery Capacity Larger batteries generally require longer charging
- Temperature Minor influence on charging duration
- Typical Charging Duration 1–4 Hours Findings
- Charging rate is the most influential factor affecting charging duration.
- Vehicles requiring a larger battery gain generally take longer to charge.
- Larger battery capacities are associated with longer charging sessions.
- Environmental temperature has only a small impact compared to charging rate and battery characteristics

6. Predictive Analysis

Objective 6: Predict Charging Duration Using Machine Learning

Purpose: To build a Random Forest Regression model for predicting charging duration.

Machine Learning Model

Algorithm: Random Forest Regressor

Input Features

Battery Capacity (kWh)

Energy Consumed (kWh)

Charging Rate (kW)

Cost per kWh

Battery Gain

Vehicle Age

State of Charge (%)

Predicted Output

Charging Duration (Hours)

Model Performance

Replace these values with your actual model results.

Metric Result

MAE : 0.8812639382273617

RMSE : 1.1030576323622425

Findings

- The Random Forest model predicts charging duration using charging session characteristics with good accuracy.
- Charging rate, battery gain, and energy consumed are the most important features for prediction.
- The predicted charging duration is used as the primary input for the AI Smart Charging Advisor.
- The machine learning model enables proactive charging recommendations instead of relying solely on historical analysis.

15.Conclusion

This project successfully analyzed electric vehicle (EV) charging patterns using exploratory data analysis, statistical analysis, and machine learning. The analysis showed that most charging sessions cost between

A Random Forest Regression model was developed to predict charging duration based on charging session characteristics. Using these predictions, an AI-Powered Smart EV Charging Advisor was designed to recommend the best charging station by considering charging duration, charging cost, and charging rate.

Overall, this project demonstrates how data analytics, machine learning, and AI can support smarter and more cost-effective EV charging decisions, helping both EV users and charging station operators improve efficiency and enhance the overall charging experience.
